A Multi-Module Limit-Order-Book Trading System

Team 1 | QuantiFi 2026 — Leena Li, Charles Zhou, Su Pu, Chloe Gao — code file `Team01QuantiFi2026.py`

Summary. Our strategy is a modular per-product portfolio that actively trades 18 of the 26 products with modules from five strategy families: market making, voucher implied-volatility smile relative value, EMA-anchored z-score mean reversion, trend following, and blended fixed-centre / adaptive-anchor mean reversion. On the official Portal backtest it earned **523,191.5 COINS** (225,130 / 188,395 / 109,666.5 on days −2 / −1 / 0), with largest portfolio intraday drawdowns of 15,404.5 / 14,445.5 / 35,672 COINS. All results are in-sample.

1. Architecture

Every product group gets its own causal module that turns order-book data into a **target position**; a shared execution layer converts targets into limit-safe orders (with in-flight order reservation for every module except `GOLDEN_PEARLS`, which has no cross-tick pending-order management), and a risk layer adds a drawdown kill switch on the trend products (Figure 1).

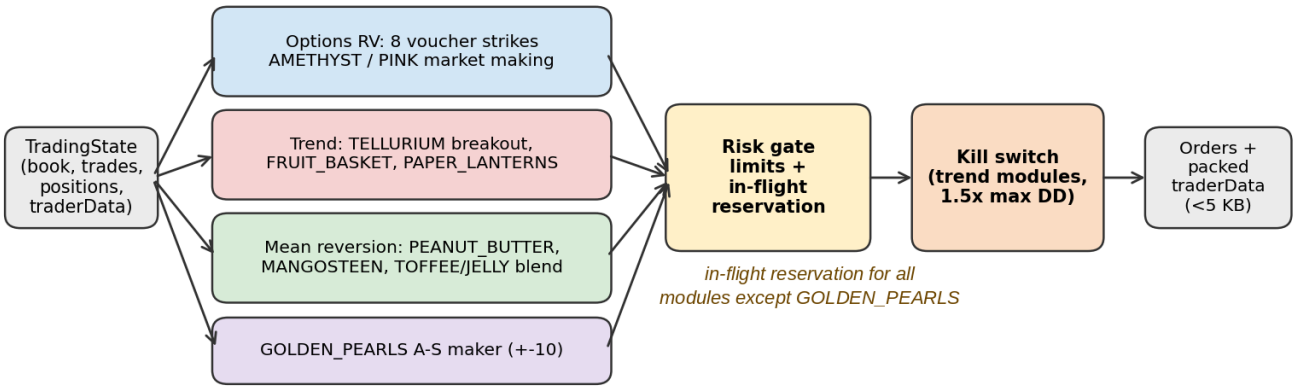


Figure 1. Our strategy: per-product modules, shared execution and risk layers.

2. Signal Mechanisms

2.1 Evidence for the family assignment

On the Portal order books we measured the average quoted spread, the per-tick volatility of mid-price changes σ_1, the lag-1 autocorrelation of mid changes ρ_1, the 100-tick variance ratio $VR_{100} = \mathrm{Var}(\Delta_{100})/(100\cdot\mathrm{Var}(\Delta_1))$ (below 1 indicates reversion, above 1 persistence; it describes only the 100-tick scale and is a diagnostic, not on its own sufficient to decide a product's family) and the average intraday high–low range (Table 1, averages over the three days).

Product	Spread	σ_1	ρ_1	VR_{100}	Range	Family
AMETHYST	3.0	0.50	−0.50	0.01	1	Market making
PINK_PEARLS	4.0	1.00	−0.13	0.76	194	Market making (implied fair)
GOLDEN_PEARLS	3.0	0.52	−0.42	2.26	84	Market making (inventory skew)
TOFFEE_CAKE	14.1	3.10	−0.23	0.45	138	Blended MR
JELLY	11.0	3.13	−0.04	1.09	363	Blended MR
PEANUT_BUTTER	6.0	4.79	−0.01	1.02	574	EMA z-score MR
MANGOSTEEN	4.0	2.48	−0.03	0.88	306	EMA z-score MR
PAPER_LANTERNS	4.0	1.51	−0.50	0.15	181	Trend (rolling line)
TELLURIUM	9.7	5.84	−0.01	1.22	1,185	Trend (swing breakout)
FRUIT_BASKET	6.0	7.95	0.00	0.97	1,744	Trend (EMA crossover)

Table 1. Order-book statistics from the portal backtest (prices in COINS; Δ_n = n-tick mid change).

`AMETHYST` is pinned (range 1, VR_{100} = 0.01), so the only edge is the spread. `TOFFEE_CAKE` pairs the widest spread with clear reversion (ρ_1 = −0.23, VR_{100} = 0.45). `TELLURIUM` and `FRUIT_BASKET` have the largest volatility and ranges (1,185 and 1,744 COINS), so holding through a drift earns far more than the half-spread paid. `PAPER_LANTERNS` bounces tick-to-tick (ρ_1 = −0.50) yet travels 181 COINS a day, hence a 50-tick regression slope. `PEANUT_BUTTER`, `MANGOSTEEN` and `JELLY` revert on the slower EMA-anchor horizon (half-life 500–3000 ticks).

Table 2 lists each module and its parameters.

Product(s)	Module	Key parameters
AMETHYST	Fixed-fair maker	fair 10000, cap 50, clip 5
PINK_PEARLS	Implied-fair passive maker	fair = 50,000 − other four mids; fair ± 2, skew 2q/20, never crosses touch, cap 20
GOLDEN_PEARLS	Avellaneda–Stoikov-style maker [2]	EMA fair (hl 20), r = fair − 2q/20, half-width max(1, σ), size 10, ±10 budget
VOUCHER_7000–7900 (8)	Smile relative value	thr 7.0, min 4 fitted strikes, r = 0, sweeps levels ≥ 7 through fair, cap 50 each
PEANUT_BUTTER	EMA z-score MR, frozen-anchor exit	anchor_hl 500, sd_hl 1000, warm-up 300, z 2, max hold 1000, cap 50
MANGOSTEEN	EMA z-score MR	hl 500, z 2, cap 40
PAPER_LANTERNS	Rolling-line trend with hysteresis	window 50, entry 0.05, exit 0.02, fair gate on, forecast horizon 50, cap 50
TELLURIUM	Swing-breakout trend	threshold 10, cap 40
FRUIT_BASKET	Own-price EMA-crossover trend	fast hl 50, slow hl 200, entry band = half-spread, warm-up 500, cap 40
TOFFEE_CAKE, JELLY	Blended MR	5000 ± 40 / 2000 ± 80; EMA hl 3000, z 2; cap 25 + 25; best level only

Table 2. Modules and key parameters (caps in lots; hl = EMA half-life in ticks).

2.2 Market making

`AMETHYST` and the pearls trade around values that are constant or recoverable from related books, so the edge is providing liquidity. `PINK_PEARLS` uses fair = 50,000 minus the other four pearls' mids (Section 5), quotes passively at fair ± 2 with skew 2q/20 and never crosses the touch. `GOLDEN_PEARLS` uses an EMA fair value (half-life 20) and an inventory-shifted reservation price in the spirit of Avellaneda–Stoikov [2], r = fair − 2q/20 (skew 2; 20 = official limit; the γqσ² term is inactive, γ = 0), quoted at half-width max(1, σ), size 10, within an outer ±10 inventory budget and with no cross-tick pending management.

2.3 Voucher smile relative value

Voucher prices across strikes should lie on one smooth volatility smile; a strike that deviates from it is likely mispriced. Using the Black–Scholes call [1] with r = 0 written in total volatility w = σ√T,

$$C(S,K,w) = S\,N(d_1) - K\,N(d_1 - w), \qquad d_1 = \frac{\ln(S/K) + w^2/2}{w}$$

we solve each strike's implied w_k by safeguarded Newton, then fit a least-squares quadratic in moneyness x = ln(K/S), **leaving the target strike out**:

$$\hat{w}(x) = a + b\,x + c\,x^2, \qquad \text{fair}_k = C(S,K,\hat{w}_{-k}(x_k))$$

We buy when fair_k − ask ≥ 7.0 and sell when bid − fair_k ≥ 7.0, sweeping every book level that qualifies. Leaving the target out stops a mispriced strike from pulling the smile toward itself. A strike is traded only if at least four other strikes have a valid implied volatility; with r = 0 the price depends on T only through w, so only current quotes are needed (Figure 2). All eight interior strikes reached their 50-lot cap on every day.

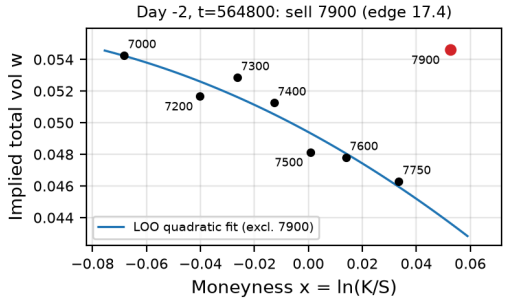


Figure 2. One portal tick (day −2): implied total volatility w by strike, the quadratic smile fitted without VOUCHER_7900, and the 7900 strike (red) sitting above it; our strategy sold it (edge 17.4 vs threshold 7.0). Only strikes 7000–7900 are shown; valid wing quotes (VOUCHER_6800/8100) are also used in the fit.

2.4 Lagged-anchor mean reversion

`PEANUT_BUTTER` and `MANGOSTEEN` oscillate around a slowly moving level. We measure deviation from a lagged EMA anchor and scale it by an EWMA volatility (for `PEANUT_BUTTER`, anchor half-life 500 ticks and volatility half-life 1000 ticks):

$$d_t = \text{mid}_t - \text{EMA}_{t-1}(\text{mid}), \qquad z_t = \frac{d_t}{\sqrt{\text{EWMA}(d^2)}}, \qquad \text{enter when } |z_t| \geq 2$$

For PEANUT_BUTTER the exit target is the anchor frozen at entry, so a drifting EMA cannot move the take-profit level away from the trade, and a maximum holding time caps exposure if reversion fails (Figure 3). No stop-loss is used: on PEANUT_BUTTER every stop variant we tested had a negative net.

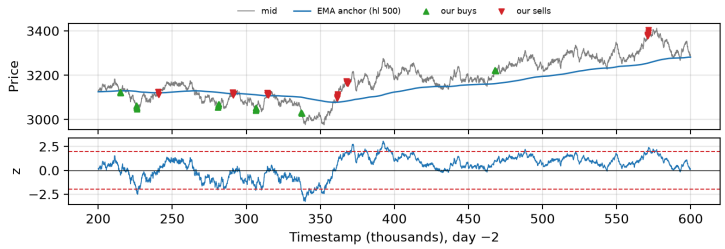


Figure 3. PEANUT_BUTTER, day −2: mid price, lagged EMA anchor and our fills (top); z-score with the ±2 entry bands (bottom).

2.5 Trend following

PAPER_LANTERNS, TELLURIUM and FRUIT_BASKET show persistent intraday drifts that mean-reversion modules would fight. PAPER_LANTERNS uses the slope of a rolling linear fit with separate entry and exit bands (hysteresis), avoiding flips around a single threshold. A fair gate filters entries: the forecast exit level is the fitted value now plus 50 ticks of slope, and a level is taken only if this forecast beats the price by more than half the current spread. TELLURIUM enters on a break of the last confirmed swing high or low, using only already-confirmed pivots, so no future data is used. FRUIT_BASKET trades its own mid with an EMA crossover: $d_t = \text{EMA}_{50} - \text{EMA}_{200}$ (half-lives in ticks, both from prior ticks). After a 500-tick warm-up it targets +40 lots when d_t exceeds half the bid–ask spread and −40 when it is below minus half the spread, and goes flat once d_t changes sign against the position. Figure 4 shows the two illustrated modules.

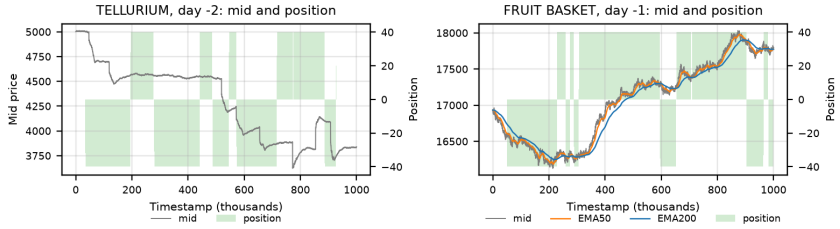


Figure 4. Trend modules on the portal data: TELLURIUM swing-breakout position (day −2) and FRUIT_BASKET EMA-crossover position (day −1), each against its mid price.

2.6 Blended mean reversion

TOFFEE_CAKE and JELLY revert around stable centres on the observed days, but a fixed centre is fragile if the level shifts. Each cap is split into two 25-lot legs: a fixed-centre leg trading 5000 ± 40 (TOFFEE_CAKE) or 2000 ± 80 (JELLY), and an adaptive leg trading $|z| \geq 2$ around an EMA anchor (half-life 3000). We shifted price levels (±100 TOFFEE_CAKE, ±200 JELLY) and added jumps and drifts (Figure 5). A fixed centre loses its edge under a level shift (worst case −6,496); an adaptive EMA3000 centre is robust (worst −1,105) but keeps only 74% of normal profit; the 25 + 25 blend keeps 92% (72.8k vs 78.9k, two days, earlier development run; the final version, with different code and execution, earns 39,063 + 37,526 = 76,589) with a worst case of −3,419. Stop-losses made every variant worse, so there are none. Trading at the best level only saved 7.8k in spread and added 6.1k net over three days.

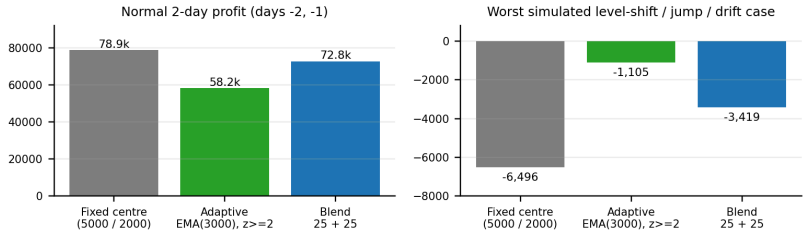


Figure 5. Fixed, adaptive and blended centres under normal days and simulated level shifts, jumps and drifts.

2.7 Execution

A shared helper converts the gap between target and position into orders and reserves capacity for orders that may still be working on the same side (R_{buy}, R_{sell}), so the limit L is never breached even when fills arrive late (`GOLDEN_PEARLS` is the exception: it has no cross-tick pending-order management and relies on its ±10 inventory budget). Orders live one tick, so the cross-tick reservation is a conservative choice that can waste some capacity:

$$Q_{buy} \le \max(0, L - q - R_{buy}), \quad Q_{sell} \le \max(0, L + q - R_{sell}).$$

Options sweep every book level at least 7 COINS through fair value; all other directional modules trade at the best level only, and the three makers rest passive quotes. Of 67,548 lots traded on the Portal (7,442 fills), 97.8% were aggressive; the makers' 1,504 lots were all passive and filled 1.51 (`AMETHYST`), 1.12 (`GOLDEN_PEARLS`) and 0.06 (`PINK_PEARLS`) COINS better than mid. Directional products touched their caps every day, while the makers peaked at 38 / 17 / 8 lots, well inside their limits.

3. Risk Management

3.1 Intraday drawdown kill switch

The switch covers the three trend products. It tracks each product's intraday PnL from our own fills (positions marked at mid) and its running peak:

$$\mathrm{PnL}_t = \mathrm{cash}_t + \mathrm{pos}_t \times \mathrm{mid}_t, \quad \mathrm{peak}_t = \max_{s \le t} \mathrm{PnL}_s, \quad \text{trigger if } \mathrm{peak}_t - \mathrm{PnL}_t > \theta_p$$

The kill-switch thresholds θ_p are 20,275.5 / 40,680 / 765 COINS for `TELLURIUM` / `FRUIT_BASKET` / `PAPER_LANTERNS`, 1.5× the largest intraday drawdown on development days −2 and −1. Once triggered, the product is flattened across the touch and receives no more orders that day; the flatten size subtracts possibly-working same-side orders, so it never crosses zero:

$$\mathrm{qty}_{\mathrm{flatten}} = \max\left(0,\ |\mathrm{pos}_t| - (\text{same-side orders still working})\right)$$

We compared multipliers 0.5×–3× and graded variants (halve, then stop) by two costs: profit missed when the price rebounds versus losses saved when the move continues (Figure 6). In a synthetic development experiment on an early prototype (60 stress days; not an independent validation of the final version) we chose 1.5× to balance retained historical profit and synthetic downside, rather than to maximize net benefit (1.5×: mean −5,132, worst day −135,696; 2.0×: mean −4,427, worst day −145,745). It did not fire on any of the three real days; 1.25× would have cut about 18k of real profit on day 0, and tighter thresholds 45k–187k. Graded protection halves the false-trigger cost but saves less, so its net is lower. In synthetic stress the switch cut the average daily loss from 44k to 5k and the worst-day loss from 437k to 136k. Mean-reversion modules deliberately have no kill switch: on `PEANUT_BUTTER` every stop variant had a negative net.

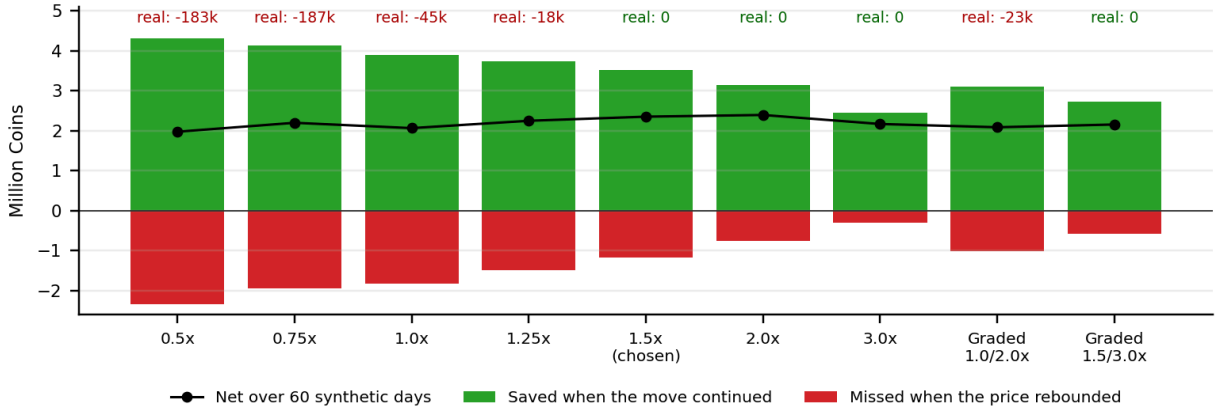


Figure 6. Kill-switch calibration by multiplier: missed rebound profit vs saved loss. 'real' labels show historical profit forgone versus no guard across days −2, −1 and 0; k denotes 1,000 COINS.

3.2 Execution safety and safeguards

With the switch forced to fire under 1–3 tick order delays and duplicate callbacks, the flatten never crosses zero or reopens, and each own fill is booked once (de-duplicated by timestamp).

Table 3. Safeguards and residual exposure.

Safeguard	What it controls	What remains exposed
Hard position caps (10–50 lots) + order reservation (not GOLDEN_PEARLS)	Limit breaches from late or pending fills	Full-cap exposure in a regime break
Kill switch (trend products)	Intraday loss after a trend reversal	Realized loss can exceed the threshold when depth is thin
Blended centres (TOFFEE_CAKE, JELLY)	Level shifts away from the fixed centre	Half the cap still on the fixed leg
Duplicate-callback guard, fill de-duplication	Double-counting state and own fills	Not assessed
Compact `traderData`, per-module `try/except`	State loss; one failure stopping all products	A failing module stops trading its product

4. Results

All 18 traded products are positive; 53 of 54 product-days are positive (the only loss is `MANGOSTEEN` on day −2, −538 COINS). Daily PnL falls from day −2 to day 0, mainly in the trend family, from 96,222 to 29,884 COINS (Figure 7).

Table 4. Portal PnL by family, days −2 to 0.

Family	PnL (COINS)	Share
Trend following	217,532	41.6%
Voucher smile RV	120,129	23.0%
Blended MR	104,202	19.9%
EMA-anchor MR	76,259	14.6%
Market making	5,069.5	1.0%
Total	**523,191.5**	100%

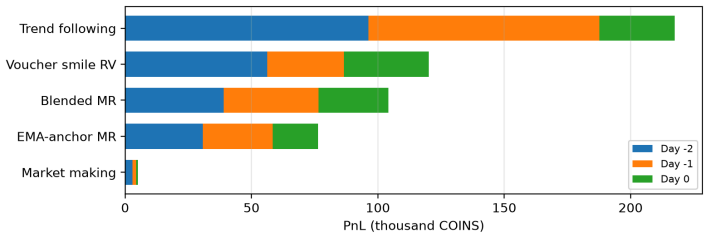


Figure 7. PnL (thousand COINS) by strategy family and day.

5. Breadth: Untraded Products

- **Pearls.** The five pearls' mids sum to about 50,000 (s.d. about 5 on both development days). WHITE/BLACK/BLUE moves offset each other within the same tick, with follow-through under 1 point against spreads of 4–8; none of 13 standalone models passed validation. We exploit the relation on `PINK_PEARLS` instead: fair = 50,000 minus the other four mids cuts its residual s.d. from 41–56 to about 5.
- **LYCHEE** is the option model's spot input. Voucher flow leads it by about 1.1 points, below the ~2-point half-spread.
- **DRAGONFRUIT / GRANADILLA.** Trend profit came from few episodes; market making failed at one-tick delay.
- **VOUCHER_6800 / 8100.** Poor wing fit and losses on development days; they still inform the smile fit.

6. Local Backtester Alignment

We aligned our local backtester with the official Portal (days −2/−1/0 summed, one-tick order lifetime, the `traderData` size cap and fill de-duplication), after which our local replay matched the Portal exactly.

7. Limitations and Next Steps

All results are in-sample (all three days were reused); `TELLURIUM` is parameter-sensitive (thresholds 10 / 20 / 40 gave 95k / 14k / 46k over three days), one reason the kill switch covers it. Candidates were also replayed pessimistically (strict passive fills, half passive volume, 1 COIN per lot).

Table 5. Main exposures and next steps.

Exposure	Why	Next step
Trend concentration (41.6% of PnL)	Edge needs persistent drifts; trend PnL fell 96,222 → 29,884	Scale kill-switch thresholds and entries to rolling volatility
Parameter selection on three days	TELLURIUM threshold, basket model chosen in-sample	Validate on new days; prefer flatter parameter regions
Spread cost	97.8% of lots cross the spread	Passive entry where the signal horizon allows
Unhedged vouchers	Eight strikes at cap move together with LYCHEE	Delta-hedge with LYCHEE; revisit wing strikes
Fixed thresholds and centres	Kill switch may fire early/late; fixed legs fragile to level shifts	Volatility-scaled bands; adaptive weight on the blend

References

[1] F. Black and M. Scholes, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, 81(3), 637–654, 1973.
[2] M. Avellaneda and S. Stoikov, "High-frequency trading in a limit order book," *Quantitative Finance*, 8(3), 217–224, 2008.